

PXE Boot on UniFi — Research Findings & Implementation Log

Date: 2026-06-22 (session 2), original 2026-06-19 **Author:** Claude (Juniper AI) **Status:** ●
BLOCKED — iPXE shim is CA 2011 only; CA 2011 removed from IMAGE-ME's Secure Boot db

Immediate fix: Disable Secure Boot on IMAGE-ME in BIOS (F1 → Security → Secure Boot → Disabled). No DHCP changes needed. Long-term fix: wait for a CA 2023-signed iPXE shim (iPXE team working on it, [rhboot/shim-review#319](#)).

2026-08-06 — Lenovo 720S-15IKB (81AC) imaging quirks + fleet self-heal fixes

- **Secure Boot must be re-enabled AFTER imaging.** The 720S-15IKB (type 81AC) only boots the imaging USB with Secure Boot **OFF**, but leaving it off in Windows breaks the Lenovo UEFI firmware capsule update (Windows Update; Device Manager → Firmware → "Lenovo UEFI System Firmware ..." shows **Code 10, "Firmware update failed due to system authentication error"**) and makes the `secure-boot` (CA-2023 enrollment) orchestrator phase spin its full 6 rounds and give up. The orchestrator cannot flip Secure Boot (physical BIOS setting) — re-enable it in BIOS once Windows is installed (the MS-signed Windows Boot Manager boots fine with SB on; only the imaging media needed it off).
 - **Driver discovery is Win10-aware for 81AC.** `discover-drivers.ps1` now evaluates both Win10 + Win11 catalogs; 81AC's real drivers only exist under Win10. See `docs/driver-library.md`.
 - **Network self-heal on connect.** `orchestrator.ps1`'s `JuniperNetworkPrivate` task now also triggers on NetworkProfile/Operational **event 10000** (network connected, 5 s delay), not just startup/logon — so Wi-Fi that associates a few seconds into boot, reconnects at the user's desk, or comes up before any user logs in gets re-asserted **Private**. (Public left WinRM/SMB firewalled off cross-subnet — the recurring "imaged machine lands Public" defect.)
-

⚠ CURRENT BLOCKER: MS UEFI CA 2011 is expiring / removed from ThinkPad db

Microsoft Corporation UEFI CA 2011 — the certificate used to sign ALL third-party UEFI boot applications including every Linux shim and the iPXE shim — **expires June 26, 2026** (this week).

Lenovo pushed ThinkPad firmware updates proactively removing CA 2011 from the Secure Boot db and replacing it with CA 2023. IMAGE-ME (ThinkPad P14s Gen 5) has received this firmware

update.

What's signed by what

Component	Signing	IMAGE-ME result
<code>ipxe-shim.efi</code> (iPXE v2.0.0)	MS UEFI CA 2011 only (1 cert in PE table)	❌ Rejected — CA 2011 not trusted
<code>wimboot</code> (v2.9.0)	CA 2011 + CA 2023 (dual-signed, 2 certs in PE table)	✅ Would pass Secure Boot
<code>EFI/Boot/bootx64.efi</code> (<code>bootmgfw_EX.efi</code>)	Windows PCA 2011 (separate chain from UEFI CA)	✅ Passes (Windows chain still trusted)
CA 2023-signed iPXE shim	Does not exist yet — iPXE team tracking in <code>rhboot/shim-review#319</code>	—

Key finding: wimboot IS dual-signed and would pass IMAGE-ME's Secure Boot. But wimboot requires iPXE to inject `BCD/boot.sdi/boot.wim` into memory — it cannot function as a standalone PXE boot file. So the chain still needs a working iPXE shim, which requires CA 2023 signing.

How we confirmed dual-signing

PE certificate table analysis (not `Get-AuthenticodeSignature`, which only reads the primary cert):

```
# on pc-deploy via WinRM:
# shimx64.efi: CertTableSize=9728 (one cert) → CA 2011 only
# wimboot:    CertTableSize=19232 (two certs, 9728+9504) → CA 2011 + CA 2023
```

Why no CA 2023 iPXE shim exists yet

iPXE v2.0.0 (March 6, 2026) shim was submitted to Microsoft for signing before the October 2025 dual-signing cutover, so it only got CA 2011. The iPXE team is working on a re-submission via the shim-review process (`rhboot/shim-review#319`). No release date. The next iPXE release will include a CA 2023-signed (or dual-signed) shim.

Source: github.com/ipxe/ipxe/discussions/1638 — iPXE collaborator NiKiZe confirmed this directly in March 2026.

Enterprise Research: Secure Boot PXE in June 2026 (post-CA2011 expiry)

Research conducted 2026-06-23. Sources: Lenovo Press LP2353, TechDirectArchive 4-part series (Parts 2/3/4), miloch.dev, systemcenterdudes.com, 2Pint Software, winreflection.com.

The Two Cert Chains — Critical Distinction

There are **two completely separate Microsoft Secure Boot certificate hierarchies**. Confusing them is the most common mistake in post-CA2011 PXE troubleshooting:

Chain	Signing authority	Validates	IMAGE-ME status (after Lenovo FW update)
Microsoft Corporation UEFI CA 2011	Signs third-party UEFI boot apps	iPXE shim, Linux shims, unsigned drivers	✗ REMOVED from db
Microsoft Corporation UEFI CA 2023	Replacement for UEFI CA 2011	Same, next generation	✓ ADDED to db
Microsoft Windows Production PCA 2011	Signs Windows OS boot components	bootmgfw.efi, winload.efi, boot.wim internals	✓ STILL TRUSTED

The **Lenovo firmware update ONLY removed UEFI CA 2011**. The Windows Production PCA 2011 (which signs `bootmgfw_EX.efi`, `winload.efi`, and everything inside `boot.wim`) is **unaffected**.

This means: our existing `boot.wim` WinPE components (`winload.efi`, etc.) do **not** need to be replaced. They use the Windows Production PCA 2011 chain, which IMAGE-ME still trusts.

The **ONLY thing broken is the iPXE shim** — it is signed by UEFI CA 2011 (third-party UEFI CA), which is now removed from IMAGE-ME's db.

How Enterprises Are Solving This Right Now

Path A: WDS + Windows Server 2022 (`boot_EX` directory)

Microsoft ships CA 2023-signed WDS PXE loaders in a special directory on Windows Server 2022 only:

```
C:\Windows\System32\RemInst\boot_EX\x64\  
  wdsmgfw_EX.efi    ← CA 2023-signed WDS PXE loader  
  bootmgfw_EX.efi   ← CA 2023-signed Windows Boot Manager
```

Fix for WDS environments:

```
copy "C:\Windows\System32\RemInst\boot_EX\x64\wdsmgfw_EX.efi" "F:\RemoteInstall\Boot\x64\wdsmgfw.efi" /Y  
copy "C:\Windows\System32\RemInst\boot_EX\x64\bootmgfw_EX.efi" "F:\RemoteInstall\Boot\x64\bootmgfw.efi" /Y  
net stop WDSserver && net start WDSserver
```

Caveats: – `boot_EX` directory only exists on **Windows Server 2022** (fully patched). Does NOT exist on Windows 11. Therefore **this approach is not available for pc-deploy**. – MDT environments: when MDT rebuilds LiteTouchPE, it regenerates WinPE from the ADK baseline which still contains CA 2011 components. **MDT overwrites any manual patches** — the fix is not

persistent. – As of May 2026, the ADK servicing updates (10.1.28000.1) still contain outdated WinPE binaries. True fix requires Microsoft to ship a corrected ADK. (TechDirectArchive Part 4, May 15, 2026)

What DOES work persistently: Build a custom WinPE outside MDT/WDS (Lenovo Press LP2353 approach). In that model, MDT never touches the WIM. For us this doesn't matter — we already build custom WinPE using ADK tools directly, without MDT.

Path B: ConfigMgr (SCCM) 2509 PXE Responder

Starting with SCCM ConfigMgr 2509, Microsoft added a checkbox: **"Use Windows Boot Loader signed with Windows UEFI CA 2023"**

This is the **cleanest enterprise solution** for CA 2023 Secure Boot PXE. Requirements: – ConfigMgr 2509 or later – ConfigMgr PXE Responder (NOT WDS — ConfigMgr has its own PXE service) – Latest ADK + WinPE Add-on

This is a Windows Server + SCCM-licensed solution. Not applicable to our Windows 11 setup.

Path C: iPXE + wimboot (Our Approach) — Waiting for CA 2023 Shim

Our stack (tftpd64 → iPXE shim → iPXE → wimboot → boot.wim via HTTP) is architecturally correct and actually **superior** to WDS for our use case:

Feature	WDS approach	iPXE + wimboot approach
boot.wim delivery	TFTP (requires WDS ramdisk extensions)	HTTP (no WDS needed)
Works on Windows 11 server	❌ WDS is Windows Server-only	✅ Works on Windows 11
MDT regeneration problem	❌ MDT overwrites patches	✅ No MDT used
Secure Boot (right now)	⚠️ Complex, requires boot_EX + custom WinPE	🔄 Waiting for CA 2023 shim
WiFi PXE support	❌ TFTP too slow/large for 802.11	✅ HTTP scales fine

The one missing piece: a CA 2023-signed iPXE shim (UEFI CA 2023). Issue tracked: – rhboot/shim-review#319 — iPXE shim submission – github.com/ipxe/ipxe/discussions/1638 — iPXE collaborator NiKiZe confirmed CA 2023 shim in progress

Once that shim is released: drop it into `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi`, re-enable Secure Boot on target machines. No other changes required.

Why the 0xc0000225 Error Happens Without WDS

When using `bootmgfw.efi` directly (without iPXE/wimboot) as the TFTP boot file, Windows Boot Manager tries to download `boot.wim` via TFTP using WDS-specific TFTP ramdisk extensions.

Standard TFTP servers (tftpd64, SolarWinds, etc.) do NOT implement these WDS extensions. Result: 0xc0000225.

This is confirmed independently by winreflection.com (Aug 2024) — same error, same conclusion. iPXE + wimboot bypasses this entirely by delivering boot.wim via HTTP.

Why Our boot.wim Does NOT Need CA 2023 Updates

A common misconception (from MDT/WDS articles) is that `winload.efi` inside boot.wim needs to be updated to CA 2023. This is ONLY true in environments where the device has had the **Windows Production PCA 2011** revoked from its db — which is a separate, additional enforcement step.

IMAGE-ME (ThinkPad P14s Gen 5, Lenovo firmware update) had **UEFI CA 2011 removed** (the third-party UEFI CA), but **Windows Production PCA 2011 remains trusted**. Therefore:

- `winload.efi` inside boot.wim (signed by Windows Production PCA 2011) →  still trusted
- `bootmgfw_EX.efi` (Windows Production PCA 2011) →  still trusted
- Only `ipxe-shim.efi` (UEFI CA 2011) →  no longer trusted

The techdirectarchive articles describe environments where **BOTH** CA 2011 chains are being revoked (some Dell SafeBIOS devices), which is a more aggressive enforcement. That situation DOES require CA 2023 updates to WinPE internals. For IMAGE-ME, only the iPXE shim needs to change.

Sources

Source	Published	Relevance
Lenovo Press LP2353	2025-2026	Official Lenovo guide: build custom WinPE outside MDT with CA 2023
TechDirectArchive Part 2	2026-04-27	Updating WinPE boot images with CA 2023 certs
TechDirectArchive Part 3	2026-05-08	Dell SafeBIOS PXE; boot_EX source files; trust gap analysis
TechDirectArchive Part 4	2026-05-15	Key: ADK still broken as of May 2026; MDT overwrites patches; confirms no durable fix without MS ADK update
miloch.dev SCCM playbook	2026-05-21	Full enterprise playbook: AvailableUpdates registry, SCCM Baselines, event-log signals
System Center Dudes	2026-06-09	SCCM boot images for hardened Secure Boot; CA 2023 checkbox in ConfigMgr 2509
2Pint Software	2024-04-19	KB5025885 + boot WIM updates; TrustedInstaller permission issue replacing winload.efi
winreflection.com	2024-08-15	PXE without WDS on UniFi; confirms 0xc0000225 with bootmgfw.efi UEFI boot without WDS

✓ DHCP OPTIONS — already applied, no changes needed

Field	Current value	Notes
Boot server (Option 66)	192.168.5.141	Correct
Boot filename (Option 67)	ipxeboot/x86_64-sb/ipxe-shim.efi	Applied by Jason 2026-06-22
Network Boot	Enabled	Correct

Recommended Interim Solution: Disable Secure Boot for Imaging

Until a CA 2023-signed iPXE shim is released:

1. On IMAGE-ME: **Enter BIOS (F1 at Lenovo logo)** → **Security** → **Secure Boot** → **Disabled** → **Save & Exit**
2. No DHCP changes needed — `ipxeboot/x86_64-sb/ipxe-shim.efi` works with Secure Boot off
3. PXE boot IMAGE-ME: shim → iPXE → HTTP → wimboot → WinPE → `deploy.ps1`

With Secure Boot disabled, UEFI skips signature validation entirely. The existing chain runs as designed. After imaging, Secure Boot can be re-enabled in BIOS; Windows boots fine either way.

For the long-term WiFi PXE + Secure Boot goal: the stack is ready. wimboot is already CA 2023 dual-signed. bootmgfw_EX.efi is Windows-chain signed. When iPXE releases a CA 2023 shim, drop it into `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi` and re-enable Secure Boot — done.

Architecture History (this session, 2026-06-22)

What happened with the bootmgfw_EX.efi approach

A test with `ipxe-http-embedded.efi` (DHCP option 67) confirmed that the unsigned iPXE got a **Secure Boot Violation** — expected, since that binary isn't signed by any key in the ThinkPad's db.

The bootmgfw_EX.efi approach (set earlier in the session) was superseded because: — `boot.wim` (513 MB) cannot be delivered via TFTP without WDS — UEFI TFTP has a ~4 GB limit but the Windows Boot Manager tries to load it at boot time and fails with **0xc0000225** ("Required device is inaccessible") — confirmed by WinReflection article author who hit the exact same issue. — The correct solution for non-WDS servers is: **iPXE + wimboot** (wimboot loads boot.wim via HTTP).

Why we're using iPXE v2.0.0 (released March 6, 2026)

iPXE v2.0.0 added official Secure Boot support via a dedicated Microsoft-signed shim: — `ipxe-shim.efi` — signed by **Microsoft Corporation UEFI CA 2011** (in every PC's UEFI db) — The shim internally verifies and loads `ipxe.efi` (signed by iPXE's own CA) — No MOK enrollment required — the shim chain is fully trusted out of the box — wimboot is already signed by Microsoft —

passes Secure Boot validation - `autoexec.ipxe` controls boot behavior (TFTP-fetched by iPXE, same directory as shim)

This is the correct long-term path. It also supports WiFi PXE (802.11 clients) once configured, because wimboot + UEFI HTTP avoids the TFTP size limitation entirely.

Files deployed to pc-deploy (2026-06-22)

Path on pc-deploy	Description	Status
C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi	MS UEFI CA 2011-signed shim, 1,038,920 bytes	✔ deployed
C:\tftpd64\ipxeboot\x86_64-sb\ipxe.efi	iPXE binary (signed by iPXE CA), 397,160 bytes	✔ deployed
C:\tftpd64\ipxeboot\x86_64-sb\shimx64.efi	Same as ipxe-shim.efi (original name in archive)	✔ deployed
C:\tftpd64\ipxeboot\x86_64-sb\autoexec.ipxe	iPXE boot script (copy of root autoexec.ipxe)	✔ deployed
C:\tftpd64\wimboot	MS-signed wimboot loader, 76,064 bytes	✔ pre-existing
C:\tftpd64\sources\boot.wim	WinPE (~513 MB)	✔ pre-existing
C:\tftpd64\EFI\Microsoft\Boot\BCD	UEFI BCD	✔ pre-existing
C:\tftpd64\Boot\boot.sdi	Boot SDI	✔ pre-existing

HTTP endpoints verified (Caddy serves C:\tftpd64)

- `http://192.168.5.141/ipxeboot/x86_64-sb/ipxe-shim.efi` → 200, 1,038,920 bytes ✔
- `http://192.168.5.141/ipxeboot/x86_64-sb/autoexec.ipxe` → 200 ✔
- `http://192.168.5.141/wimboot` → 200, 76,064 bytes ✔

autoexec.ipxe content

```
#!ipxe
echo Juniper WinPE -- Secure Boot chain (shimx64 -> ipxe -> wimboot)
set http http://192.168.5.141
kernel ${http}/wimboot
initrd ${http}/EFI/Microsoft/Boot/BCD      BCD
initrd ${http}/Boot/boot.sdi               boot.sdi
initrd ${http}/sources/boot.wim           boot.wim
boot
```

Previous state (2026-06-22, morning session)

- CA2023 IS enrolled in IMAGE-ME's UEFI db ✅ (previous "not enrolled" diagnosis was a false negative)
- bootmgfw_EX.efi IS properly CA2023-signed ✅
- PXECompatibility=0 is set on pc-deploy ✅
- BCD correctly configured ✅
- The "nope" failure from 2026-06-19 was almost certainly PXECompatibility=1 (TFTP log shows zero activity after the fix was applied at 3:18 PM that day)

Remaining uncertainty (pre-iPXE pivot): Our `bootx64.efi` is a preview/insider build (version 10.0.26100.1085, file date 4/1/2024, signing cert NotAfter 10/19/2024). A production version (3,055,456 bytes, updated 6/19/2026 by Windows Update) exists at

`C:\Windows\Boot\EFI_EX\bootmgfw_EX.efi` on IMAGE-ME. If the preview file's Authenticode hash is in the DBX, Secure Boot will still fail. This issue is moot — we've pivoted to iPXE v2.0.0 which doesn't use `bootmgfw_EX.efi`.

Network Topology

```
EFG Router / DHCP server (192.168.0.1)
├─ UniFi Core Switch (USW Pro 48 PoE)
│   └─ 5x UniFi U7 Pro XG APs
│       └─ Port 21 → TP-Link TL-SG108-M2 (Engineering – unmanaged)
│           ├── pc-deploy (192.168.5.141) ← Caddy HTTPS/HTTP, tftpd64 TFTP
│           ├── ThinkPad P14s Gen 5 (192.168.11.24) ← PXE client
│           └─ Yoga 7 2-in-1 (192.168.11.3) ← PXE client (WiFi only)
```

IP space: UniFi Default network is **192.168.0.1/20** — one flat /20 covering 192.168.0.0–192.168.15.255. All machines are on this single DHCP scope. No VLANs. No DHCP relay.

Windows Firewall on pc-deploy: DISABLED. All ports open.

Current Boot Architecture (updated 2026-06-19)

Shim has been removed. UEFI firmware now loads Windows Boot Manager directly from TFTP. This is the cleanest possible chain and is what WDS/SCCM use when Windows UEFI CA 2023 is enrolled.


```
Target UEFI firmware (Secure Boot ON)
| selects network boot entry
↓
DHCP → Option 66 = 192.168.5.141 (TFTP server)
      Option 67 = EFI/Boot/bootx64.efi (TFTP filename)
↓
TFTP: downloads EFI/Boot/bootx64.efi from 192.168.5.141:69
↓
bootx64.efi (= bootmgfw_EX.efi, Windows Boot Manager, signed by Windows UEFI CA 2023, 2,692 KB)
| verified by UEFI firmware against Windows UEFI CA 2023 in db
| IF CA2023 in db → PASSES ✓
| IF CA2023 NOT in db → Secure Boot Violation ← suspected current failure
↓
bootmgfw_EX.efi reads BCD from:
  TFTP path: EFI\Microsoft\Boot\BCD
↓
BCD: ramdisksdidevice=boot, path=\windows\system32\boot\winload.efi
     device=ramdisk=[boot]\sources\boot.wim
↓
winload.efi [⚠ currently PCA2011-signed inside boot.wim – may be a secondary issue]
↓
boot.wim (~513 MB) served via TFTP
↓
WinPE boots → deploy.ps1 → Juniper PC Deployment System
```

⚠ Known issue: winload.efi in boot.wim is PCA2011-signed

When boot.wim was inspected (2026-06-19): - `Windows\System32\boot\winload.efi` → **CN=Microsoft Windows Production PCA 2011**, 3,017 KB - `Windows\Boot\EFI_EX\bootmgfw_EX.efi` → **CN=Windows UEFI CA 2023** ✓ (only the boot manager has a PCA2023 variant) - No `winload_EX.efi` exists — Microsoft only provides a PCA2023 replacement for the boot manager, not the loader

KB5025885 adds **hash-based** DBX entries for specific vulnerable bootmgfw.efi versions — it does NOT broadly revoke the PCA2011 CA. Since our boot.wim's winload.efi is a different binary (ADK-built, not the vulnerable version), its hash is very likely NOT in the DBX. This should not be a problem. However, if it does fail, the fix is to rebuild boot.wim using a newer ADK (26100+) which includes updated components.

Key Files on pc-deploy (current state as of 2026-06-19)

Path	Description	Status
C:\tftpd64\shimx64.efi	Ubuntu shim, dual-signed UEFI CA 2011 + 2023, 1,048,424 bytes	✓ correct
C:\tftpd64\grubx64.efi	bootmgfw_EX.efi (Windows UEFI CA 2023, NotAfter 2024-10-19, 2,692 KB)	✓ PCA2023
C:\tftpd64\grubx64-ipxe-20260619.efi	Backup of old iPXE binary (Juniper CA)	📁 backup
C:\tftpd64\grubx64-pca2011-20260619.efi	Backup of PCA2011 bootmgfw.efi (NotAfter 2026-10-17)	📁 backup
C:\tftpd64\EFI\Boot\bootx64.efi	Same bootmgfw_EX.efi (Windows UEFI CA 2023, NotAfter 2024-10-19)	✓ PCA2023
C:\tftpd64\EFI\Microsoft\Boot\BCD	UEFI BCD: winload.efi + ramdisksdidevice=boot + block size opts	✓ correct
C:\tftpd64\Boot\BCD	Legacy BCD: winload.exe + ramdisksdidevice=boot	✓ correct
C:\tftpd64\sources\boot.wim	WinPE (NIC drivers injected, ~513 MB)	✓ correct
C:\tftpd64\tftpd64.ini	PXECompatibility=0, BaseDirectory=C:\tftpd64, Services=1	✓ FIXED

Caddyfile (C:\caddy\Caddyfile)

Both HTTP and HTTPS serve C:\tftpd64 as the file root.

```
http://192.168.5.141 {
  log { output file C:/caddy/access.log; format json }
  file_server { root C:/tftpd64 }
}

https://192.168.5.141 {
  tls C:/caddy/server.crt C:/caddy/server.key
  log { output file C:/caddy/access.log; format json }
  file_server { root C:/tftpd64 }
}
```

UniFi DHCP (current state — updated 2026-06-19)

Field	Value
Network Boot	☒ enabled
Network Boot — server	192.168.5.141
Network Boot — filename	EFI/Boot/bootx64.efi ← shim bypassed
TFTP Server	192.168.5.141

Changes Applied 2026-06-19 (Architecture Change)

All changes made to pc-deploy via WinRM (credentials from DPAPI cache).

1. Replace iPXE with Windows Boot Manager (PCA2011 — later upgraded)

```
# First attempt — PCA2011-signed (deployed, then rejected by ThinkPad DBX)
Copy-Item "C:\tftpd64\grubx64.efi" "C:\tftpd64\grubx64-ipxe-20260619.efi"
Copy-Item "C:\Windows\Boot\EFI\bootmgfw.efi" "C:\tftpd64\grubx64.efi"
# Result: 2,984 KB, Issuer = Microsoft Windows Production PCA 2011 — REJECTED (in DBX)
```

1b. Upgrade to PCA2023-signed bootmgfw_EX.efi

The ThinkPad has KB5025885 Stage 3: Microsoft Windows Production PCA 2011 fully revoked in DBX. The PCA2023-signed replacement (bootmgfw_EX.efi) was already present in our own C:\tftpd64\sources\boot.wim at Windows\Boot\EFI_EX\bootmgfw_EX.efi .

```
# Mount boot.wim read-only
New-Item -ItemType Directory "C:\bootmount" -Force
dism /Mount-Wim /WimFile:C:\tftpd64\sources\boot.wim /Index:1 /MountDir:C:\bootmount /ReadOnly

# Extract PCA2023 binary
$src = "C:\bootmount\Windows\Boot\EFI_EX\bootmgfw_EX.efi"
Copy-Item "C:\tftpd64\grubx64.efi" "C:\tftpd64\grubx64-pca2011-20260619.efi" # backup
Copy-Item $src "C:\tftpd64\grubx64.efi" -Force
Copy-Item $src "C:\tftpd64\EFI\Boot\bootx64.efi" -Force

dism /Unmount-Wim /MountDir:C:\bootmount /Discard
# Result: 2,692 KB, Issuer = Windows UEFI CA 2023, NotAfter = 2024-10-19 ✓
```

Windows UEFI CA 2023 is in every modern UEFI db and is NOT in any DBX. The leaf cert's NotAfter (2024-10-19) is past, but Secure Boot validates against the CA, not the leaf cert expiry.

2. Fix TFTP compatibility flag

```
; tftpd64.ini (was 1, changed to 0)
PXECompatibility=0
; UEFI clients require RFC 2347 TFTP option negotiation.
; PXECompatibility=1 forces legacy mode and breaks UEFI clients.
```

3. Add TFTP block size optimizations to all BCD stores

```
$ramdiskGuid = "{7619dcc8-fafe-11d9-b411-000476eba25f}"
foreach ($bcd in @(
    "C:\tftpd64\Boot\BCD",
    "C:\tftpd64\EFI\Microsoft\Boot\BCD",
    "C:\tftpd64\EFI\EFI\Microsoft\Boot\BCD"
)) {
    bcdedit /store $bcd /set $ramdiskGuid ramdisktftpblocksize 16384
    bcdedit /store $bcd /set $ramdiskGuid ramdisktftpwindowsize 4
}
```

4. Restarted tftpd64 service

tftpd64 was restarted to pick up the `PXECompatibility=0` change.

Why the Previous Architecture Failed

Root cause: wrong boot chain for Secure Boot + Windows imaging

The previous chain was: `shimx64.efi` → `grubx64.efi` (iPXE signed by Juniper CA)

Why this fails with Secure Boot ON

Shim's verification chain checks (in order): 1. UEFI `db` — list of trusted keys embedded in firmware 2. `MokList` — Machine Owner Keys enrolled per-machine via MokManager 3. Shim's embedded vendor cert — Ubuntu's Canonical cert, baked into `shimx64.efi`

The Juniper PXE CA cert is **none of these**. It's not in any UEFI `db`, it's not in `MokList` unless enrolled manually, and Ubuntu shim has no knowledge of it. Result: every boot attempt produced a `Secure Boot Violation` before iPXE could run.

Why MOK enrollment didn't stick

Multiple attempts were made to enroll the Juniper PXE CA via MokManager (`mmx64.efi`). All failed: ThinkShield on the P14s Gen 5 appears to wipe `MokNew` between reboots when the enrollment isn't completed within the same BIOS session. A "Configuration change. Restarting." message cleared the pending enrollment each time.

Even if MOK enrollment worked, it would require manual BIOS interaction on every machine before imaging — not viable for fleet deployment.

Why the fix works

`bootmgfw.efi` (Windows Boot Manager) is signed by `Microsoft Windows Production PCA 2011`. This certificate is **already in the UEFI db** on every PC manufactured in the last decade — it's how Windows itself boots with Secure Boot enabled. Shim loads `grubx64.efi`, verifies it against the UEFI db, finds the MS PCA 2011 cert, and passes it without any MOK enrollment.

No per-machine configuration required. This is the same chain WDS/SCCM/MDT use in every enterprise Windows deployment — the components just come from the free ADK instead of Windows Server.

Previous Root Cause: HTTPS URL as TFTP Filename (Resolved 2026-06-19)

An earlier root cause was identified and fixed:

The EFG DHCP `dhcpd_boot_filename` was set to `https://192.168.5.141/shimx64.efi` (a full URL). A PXEBOOT client only speaks TFTP — it sent that literal string as the TFTP filename. `tftpd64` received an RRQ for `https:\192.168.5.141\shimx64.efi`, which is an invalid Windows path. It silently discarded the request. Zero `tftpd64` log entries. No TFTP response.

Fix applied: Changed to `shimx64.efi` (plain filename). PXEBOOT clients now request the correct TFTP file, which exists at `C:\tftpd64\shimx64.efi`.

Confirmed via pktmon: 10 TFTP RRQ packets (76 bytes each, two source ports) visible arriving at `pc-deploy:69` from ThinkPad 192.168.11.24 (MAC C8-53-09-2D-0D-56) with zero outbound response — consistent with `tftpd64` receiving an unparseable URL-format filename.

Testing Instructions

Next test (when ThinkPad is powered on)

Boot the ThinkPad P14s Gen 5 to the F12 boot menu and select the PXE/Network Boot entry (Boot0024).

Expected success path: 1. TFTP downloads `EFI/Boot/bootx64.efi` (= `bootmgfw_EX.efi`, PCA2023) 2. UEFI validates PCA2023 → if Windows UEFI CA 2023 in db → PASSES 3. Windows Boot Manager reads BCD → mounts `boot.wim` → WinPE loads 4. `deploy.ps1` launches — ThinkPad shows the Juniper imaging prompt

If it still shows Secure Boot Violation: Claude will check via WinRM whether Windows UEFI CA 2023 is enrolled in the ThinkPad's UEFI db (`Get-SecureBootUEFI -Name db`). If it's absent, the ThinkPad needs Windows Update (KB5025885) to enroll it — Claude will trigger this remotely.

ThinkPad WinRM target: 192.168.11.24 , MAC C8-53-09-2D-0D-56 , creds from lPassword.

What success looks like

The ThinkPad displays the Windows Boot Manager loading screen (spinning dots), then WinPE loads and the deploy prompt appears.

Secure Boot Violation — Diagnostic Log (2026-06-19)

Attempt 1: PCA2011-signed bootmgfw.efi as grubx64.efi

Copied C:\Windows\Boot\EFI\bootmgfw.efi (PCA2011, 2,984 KB) as grubx64.efi . **Result:** Secure Boot Violation — "Invalid signature detected." **Diagnosis:** ThinkPad KB5025885 Stage 3 has hash-based DBX entries for this specific binary. ✓ confirmed.

Attempt 2: PCA2023-signed bootmgfw_EX.efi as grubx64.efi

Extracted bootmgfw_EX.efi (Windows UEFI CA 2023, 2,692 KB) from boot.wim\Windows\Boot\EFI_EX\ . Deployed as both C:\tftpd64\grubx64.efi and C:\tftpd64\EFI\Boot\bootx64.efi . DHCP option 67 was still shimx64.efi at this point — shim was loading grubx64.efi. **Result:** SAME Secure Boot Violation — "Invalid signature detected."

Diagnosis: Both PCA2011 and PCA2023 failing = failure is UPSTREAM of grubx64.efi, likely at shim validation level (shimx64.efi may be hash-revoked in this ThinkPad's DBX) OR Windows UEFI CA 2023 is not enrolled in the UEFI db (KB5025885 Stage 3 db enrollment hasn't run on ThinkPad).

Change applied: Bypass shim

DHCP option 67 changed from shimx64.efi → EFI/Boot/bootx64.efi (already = PCA2023 bootmgfw_EX.efi). The UEFI firmware now loads and validates bootmgfw_EX.efi directly — no shim intermediary. **PENDING TEST** (ThinkPad not powered on at time of change).

For future reference — PCA2023 source

The PCA2023 boot binary was found already inside our own boot.wim:

```
C:\tftpd64\sources\boot.wim → Windows\Boot\EFI_EX\bootmgfw_EX.efi
```

Microsoft's official script (KB5053484 / Make2023BootableMedia.ps1) copies exactly this file — there's no need to download anything external if you have a recent boot.wim.

Full Diagnostic History

pktmon Capture — 2026-06-19 (pre-fix)

```
pktmon start --etw -f C:\boot-capture.etl on pc-deploy during ThinkPad PXEBOOT attempt.
```

Traffic	Direction	Finding
192.168.0.1:67 → 255.255.255.255:68	Inbound (broadcast)	EFG DHCP ACK confirmed
192.168.11.24:1753 → 192.168.5.141:69	Inbound	5 TFTP RRQ packets (76 bytes each)
192.168.11.24:1227 → 192.168.5.141:69	Inbound	5 more TFTP RRQ (~90 seconds later)
192.168.5.141 → 192.168.11.24	Outbound	NOTHING — zero TFTP responses

10 packets, two source ports = two independent TFTP session attempts, each retried 5× on timeout. MAC `C8-53-09-2D-0D-56` = ThinkPad P14s Gen 5.

This proved: (1) EFG delivers boot options, (2) STP not blocking traffic, (3) failure on pc-deploy. Root cause: tftpd64 silently discarding RRQ with URL-format filename.

tftpd64 Diagnostic Results — 2026-06-19

Check	Result
Process	Running 
<code>netstat -ano findstr ":69"</code>	UDP 0.0.0.0:69 *.* PID — socket bound 
tftpd64.log size	0 bytes — no successful transfers (explained by invalid filename)
Windows Firewall	All three profiles OFF 
SecurityLevel	0 (no IP filtering) 
tftpd32.ini vs tftpd64.ini	Confirmed identical; tftpd64.exe reads tftpd32.ini (normal) 
PXECompatibility	Was 1 (broke UEFI TFTP option negotiation) → fixed to 0 

All Attempts Summary

What	Outcome
Standard PXEBOOT with URL as DHCP filename	TFTP arrived at pc-deploy, no response (URL filename bug)
UEFI HTTP Boot with <code>http://</code> URL	Zero Caddy hits — ThinkPad HTTPSBOOT ignores plain HTTP
UEFI HTTPS Boot (custom boot entry)	"Could not retrieve NBP file size" — TLS handshake fails
Enroll Juniper TLS CA via BIOS cert menu	No change — menu is Lenovo ON-PREMISE, not UEFI HTTPS Boot
MOK enrollment via <code>mmx64.efi</code>	ThinkShield wipes MokNew; enrollment never persists
Fixed URL-as-filename in UniFi DHCP	✅ Fixed. TFTP now requests correct filename
Fixed <code>tftpd64</code> competing DHCP (<code>Services=15→1</code>)	✅ Fixed
Replaced <code>grubx64.efi</code> with <code>bootmgfw.efi</code>	✅ Applied — PENDING TEST

Community Research Findings

Source	Platform	Finding
community.ui.com (8yr old)	USG / EdgeOS	Garbage bytes in Option 67 — not applicable to EFG/UniFi OS
community.ui.com (13yr old)	EdgeRouter	Duplicate <code>dhcpd.conf</code> entries — not applicable
kenmoini.com (2023)	UDM-Pro	Single-arch PXE works via UniFi DHCP; multi-arch needs external DHCP
HardForum (2021)	UDM	TFTP blocked by Windows Firewall — ruled out (WF disabled)
EduGeek (Dec 2025)	EFG	PXE fixed by switch reboot; another user runs Windows DHCP instead

References

- [Lenovo CDRT — ThinkPad Startup settings](#)
- [Lenovo CDRT — ThinkPad Network settings](#)
- [UEFI HTTPS Boot — tianocore wiki](#)
- [PXE & HTTP\(s\) Booting DHCP Options — hannan.au](#)
- [Ubiquiti Community — SCCM PXE Boot and UniFi](#)

- [pc-imaging-server repo](#)
-

Expected Boot Sequence (Secure Boot DISABLED — current interim path)

1. IMAGE-ME PXE boots → DHCP gives option 66= `192.168.5.141` , option 67= `ipxeboot/x86_64-sb/ipxe-shim.efi`
2. UEFI TFTP downloads `ipxe-shim.efi` (1 MB) — no Secure Boot signature check
3. Shim verifies `ipxe.efi` via its own embedded iPXE vendor cert → loads `ipxe.efi` via TFTP
4. iPXE fetches `autoexec.ipxe` via TFTP from same directory
5. iPXE executes the script: downloads `wimboot` , `BCD` , `boot.sdi` , `boot.wim` via HTTP from `http://192.168.5.141/`
6. wimboot (CA 2023 dual-signed) loads WinPE from `boot.wim` in memory
7. WinPE boots → `deploy.ps1` → Juniper imaging prompt

Expected Boot Sequence (Secure Boot ENABLED — requires CA 2023 iPXE shim, not yet released)

1. IMAGE-ME PXE boots → DHCP gives option 66= `192.168.5.141` , option 67= `ipxeboot/x86_64-sb/ipxe-shim.efi`
2. UEFI TFTP downloads `ipxe-shim.efi` (1 MB)
3. **UEFI Secure Boot validates shim against CA 2023** → PASSES once shim is CA 2023-signed
4. Shim verifies `ipxe.efi` via embedded vendor cert → loads via TFTP
5. iPXE fetches `autoexec.ipxe` via TFTP
6. iPXE downloads `wimboot`, `BCD`, `boot.sdi`, `boot.wim` via HTTP
7. **UEFI Secure Boot validates wimboot against CA 2023** → PASSES (wimboot already dual-signed)
8. wimboot loads WinPE → `deploy.ps1` → Juniper imaging prompt

When the CA 2023 shim arrives: replace `C:\tftpd64\ipxeboot\x86_64-sb\ipxe-shim.efi` and re-enable Secure Boot on target machines. No other changes needed.

Signing Evidence (collected 2026-06-22)

```
shimx64.efi (1,038,920 bytes, LastWrite 2026-03-06)
  PE CertTable: 9,728 bytes = ONE signature
  Cert 1: CN=Microsoft Windows UEFI Driver Publisher, issuer CN=Microsoft Corporation UEFI CA 2011
  → CA 2011 ONLY – fails on IMAGE-ME (CA 2011 removed from Lenovo ThinkPad firmware db)

wimboot (76,064 bytes, LastWrite 2026-06-18)
  PE CertTable: 19,232 bytes = TWO signatures
  Cert 1 (9,728 bytes): CA 2011 primary
  Cert 2 (9,504 bytes): CA 2023 secondary ← confirmed by "UEFI CA 2023" string in binary
  → DUAL-SIGNED – passes Secure Boot on IMAGE-ME ✅

bootx64.efi (bootmgfw_EX.efi, 3,055,456 bytes)
  Signer: CN=Microsoft Windows, issuer CN=Microsoft Windows Production PCA 2011
  → Windows cert chain (separate from UEFI CA chain) – IMAGE-ME trusts it ✅
```

Juniper Design internal documentation – last updated 2026-06-22 session 2

Modern imaging UI = native WPF (2026-08-07)

The on-machine imaging screen is a native PowerShell/WPF window (winpe\deploy-screen.ps1), NOT a browser. Full runbook + rationale: docs\WINPE-USB-RUNBOOK.md. Key points:

- Edge/Chromium does NOT run in stock WinPE (launches, silently exits). An Edge-kiosk experiment bloated the WIM to ~1.4 GB and never rendered; removed.
- WPF needs only WinPE-NetFx + WinPE-PowerShell (already baked). Image ~792 MB.
- Reproduces the Claude Design screen: navy chrome, Google Sans / Source Serif 4 / Roboto Mono loaded as PRIVATE fonts from X:\brand-fonts, vector wordmark (Geometry.Parse), 14 phase ticks driven by tailing X:\deploy_log.txt.
- deploy-boot.ps1 shows the WPF screen while deploy.ps1 runs hidden; console is the automatic fallback. Also baked: OpenSSH; 7 NIC/storage drivers preserved.
- Image promoted to BOTH PXE (WDS, C:\RemoteInstall) and the USB source (C:\WinPE-src), hash-verified identical; stick re-written + hash-verified.